Battery boost (TPS61288) repeated-failure debug log

Status: LEADING HYPOTHESIS (pending scope) — **a BT-channel layout asymmetry causes a destructive SW overshoot when the boost drives the bus**. FOUR battery-side TPS61288 boosts destroyed, each the moment the battery boost actively drives VBUS. The schematic is **symmetric** (every boost-stage part matches FC; only `RC` differed, now reverted; `R1` is the irrelevant ADC sense divider). So the difference is **physical/layout in the BT channel** — by elimination, the robust conclusion. Leading specific mechanism: the **output-capacitor hot loop**. All power nets are **polygon pours** (trace-length figures are irrelevant), so the metric that matters is the **Cout distance to the IC output pin**, measured directly by the operator: **FC = 40 mil, BT = 240 mil (6× farther)**. BT's larger hot loop has higher inductance → SW/VOUT rings past the **20 V abs-max** (only ~0.5–1.5 V over the 17.5 V rail) under the di/dt of driving the bus → sync-rect FET fails short (`VIN`–`SW`–`VOUT`→GND). (The SW node itself is a wide pour, 1.6× longer on BT — a minor contributor. An earlier Gerber trace-length figure of "17.45 mm / 2.2×" was **inflated/wrong** — it excluded the pours; superseded.) Fits Death 1 (original un-reworked boost — it's the copper), FC always surviving, the `RC` swap not helping, death at 120 mA (energy is internal $\frac{1}{2} \cdot L \cdot di^2$, no input limit bounds it), and the clean DC injection (no di/dt → no ring). **Confidence is now reasonably strong (6× on the dominant dimension), but confirm with one scope capture** of the SW node on bus-connect before trusting it; if BT doesn't ring past 20 V, suspect a return-path/via or manufacturing defect instead. Board-level fix: **add output ceramic right at the IC VOUT/PGND** to collapse the BT hot loop to FC-like (± SW snubber). Firmware cannot fix this. No spare boosts remain. This document is the cold-start reference — read it before touching the bench.

One-line summary: A *known-good* boost regulates 17.5 V fine standalone on the BT pad, then dies the instant it drives the bus — same as three before it, and now with the loop compensation (`RC`) matched to the always-working FC channel. So it is **not** wiring, **not** a short, **not** the supply, **not** the bring-up sequence, **not** `RC`. With `RC` matched and the only other catalogued "delta" (`R1`) now known to be the **input-voltage ADC sense divider — not a boost-loop component at all** — the two boost power stages/loops are, on paper, **identical**. Yet FC lives and BT dies. The cause is therefore **not a compensation/design value**; it is something **physical** in the BT channel — damaged output caps, a layout/pad/via defect, an *uncatalogued* component difference, or the `BT_SEQUENCE` tie — that only bites when the boost actively switches into the bus.

Key implication (Death 1): the very first death was the *original, never-reworked* BT boost. So the BT channel was hostile to a boost **before any rework** — pointing at an original manufacturing defect or layout asymmetry on the BT channel, which repeated rework may have compounded but did not create.

Pin / net reference (so names aren't confused again)

Signal	Teensy pin	Net / part	Function
`FC_REG_ENABLE`	3	`EN-REG-FC`	FC **boost** enable (TPS61288 EN)
`BT_REG_ENABLE`	4	`EN-REG-BT`	BT **boost** enable (TPS61288 EN)
`FC_BUS_ENABLE`	27	`D-FC-EN` (RT1987)	FC boost output → VBUS (ideal-diode switch)
`BT_BUS_ENABLE`	28	`D-BT-EN` (RT1987)	BT boost output → VBUS (ideal-diode switch)
`MOT_PWR_ENABLE`	29	`D-MT-EN` (RT1987)	VBUS → V-MOT / VESC

Signal	Teensy pin	Net / part	Function
`BT_SEQUENCE_ENABLE`	32	`D-BT-SQ` (RT1987)	battery (VBT) → charger VBAT terminal

Note: in earlier chat the operator said "FC_REG/BT_REG" but **meant `FC_BUS`/`BT_BUS`**. This doc uses the correct names above.

Topology facts (corrected from earlier wrong assumptions):

- VBUS carries only **~30–40 µF** (the RT1987 ceramics: `D-FC-EN` VOUT, `D-BT-EN` VOUT, `D-MT-EN` VIN, `D-BC-FC` VIN, each 10 µF, + the BUS-V divider).
- The **470 µF bulk cap is on V-MOT / regen, behind `MOT_PWR_ENABLE`** — NOT on VBUS. It was off (`MOT_PWR_ENABLE` low) in every failure. **Bus inrush is not the issue.**
- Each boost output (`VBUS-FC` / `VBUS-BT`) has **3 × 22 µF** (DC-derates to ~30 µF at 17.5 V).
- Boost: **TPS61288**, L = 2.2 µH, 15 A cycle-by-cycle switch limit, OVP 19 V (≤19.5 V), SW/VOUT abs-max 20 V, ~3 ms soft-start. (Datasheet: `references/Datasheets/TPS61288LRQQR.pdf`)

Failure datapoints

#	Source	What was done	Result
pre-1	Two 9 V batteries (FC + BT separate)	Both boosts enabled, **both bus switches OFF**	Both regulated **17.5 V** standalone, fine
pre-1	9 V batteries	Enabled `FC_BUS_ENABLE` (FC → VBUS)	FC on VBUS, **no incident**
Death 1	9 V batteries	FC already disconnected from VBUS; enabled `BT_BUS_ENABLE`	**BT boost fried**
Death 2	DC supply, **120 mA** limit, BT input only, no FC	Old FW: State-0 turned bus switches on, then boosts	Supply collapsed into CC at 120 mA; **BT boost fried**
Death 3	DC supply, **full (≥5 A)**, BT input only, no FC	New FW (BENCH_TEST): booted to Idle fine, sensors OK; sent `G` (bring-up)	Supply hit **5–7 A with collapsed voltage**; **BT boost fried**
Death 4	DC supply, **8.2 V / 200 mA limit** on BT input (board-powered), no FC	`RC-BT` reverted to **61.2 kΩ**; **known-good FC TPS61288 moved to BT pad** (regulated 17.5 V standalone after reflow); sent `G` (gentle, bus pre-charged ~7.7 V via body diode)	Draw **immediately pegged 200 mA CC**, voltage collapsed; **BT boost fried** (`VBT`↔GND short)

Other confirmed conditions:

- `BT_SEQUENCE_ENABLE` (battery → charger) was **ON in all FOUR deaths** — never once isolated OFF.
- In death 1 the FC boost was **already disconnected** when BT was connected (not a paralleling fight).
- Post-mortem each time: `VBT → GND` short (the dead boost: VIN–SW–VOUT fused to GND).
- **Death 4 conditions** (most controlled yet): `FC_BUS_ENABLE`, `MOT_PWR_ENABLE`, `REGEN_ENABLE`, `FC_CHARGE_ENABLE` all OFF; `BT_SEQUENCE_ENABLE` ON; `BT_BUS_ENABLE` brought up by `G` in sequence

with `BT_REG_ENABLE`. `G` energizes the bus switches **first**, so the bus was pre-charged to ~7.7 V (boost body diode) before the boost soft-started — **not** a 0 V hot-plug.
- **Death 4 is the decisive datapoint:** the boost was the FC channel's *proven-good* part. It lived on the FC pad, died on the BT pad. **Pad/channel, not part.**

Boost-removed path test (PASS — bus path is clean)

Teensy powered separately; **BT TPS61288 removed**; DC bench supply driving the boost-output node `VBUS-BT`; Teensy ran the bus startup (asserted `BT_BUS_ENABLE` / `D-BT-EN`). **Result: VBUS rose to 17.5 V stably, supply current ≈ 0.** So `D-BT-EN`, VBUS, and the whole BT bus path are good — no short, no low-impedance load. The fault is **not** in the path.

FC boost drives the bus (PASS — healthy reference, 2026-06-24)

Teensy powered separately; **12 V DC supply on the FC boost input**; `G` (bring-up) command. **Result: the FC boost cleanly brought VBUS up to 17.5 V.** Confirms the bring-up sequence and the separate-logic-supply rig are sound, and gives the healthy "boost actively driving the bus" reference. (BT TPS61288 still removed.)

Dual-source OR-ing (PASS — `D-BT-EN` passes a live bus cleanly, 2026-06-24)

Teensy powered separately; **9 V battery on the FC boost input** (FC boost live, holding the bus at ~17.61 V); **DC supply at 17.5 V on `VBUS-BT`** with the **BT TPS61288 still removed**; `G` command. **Result:** the bus came up cleanly to the shared voltage, then tracked **the higher of the two sources**. Sweeping the DC supply 17.5 → 17.67 V, VBUS followed smoothly 17.61 → 17.67 V — the two ideal-diode paths OR cleanly, no fight, no instability. This further confirms the BT path is clean even with `D-BT-EN` carrying current alongside a live FC source on the bus; **only the BT boost itself is implicated.**

Ruled OUT (with evidence)

- **Boost is defective / wrong part / FB-droop misconfig / inductor** — NO. Both boosts regulate 17.5 V standalone. The boost circuit works until connected to VBUS.
- **Supply collapse / inrush / motorboating / overshoot (the earlier theories)** — NOT the core cause. A stiff ≥5 A supply killed it as fast as 120 mA. Bus is ~40 µF (470 µF is elsewhere), so inrush is negligible. These transient theories are **superseded**.
- **`D-BT-EN` EXP(CD)-to-GND or VOUT-to-GND short** — NO. Ohmmeter, board unpowered: `D-BT-EN` EXP→GND, `D-BT-EN` VOUT→GND, `D-FC-EN` EXP→GND, `D-FC-EN` VOUT→GND **all open**.
- **VBUS shorted to GND** — NO. The FC boost held VBUS at 17.5 V in death 1.
- **An input current limit makes a boost test safe** — NO. Death 2 fried the boost at 120 mA. Do not rely on any input current limit to protect a boost (see Safety below).
- **A dynamic low-impedance fault / short in the BT→VBUS path** — NO. The boost-removed path test (above) drove the node to 17.5 V at ~0 current. The path is clean.
- **Charger / `BT_SEQUENCE` static load** — effectively ruled out by the path test (~0 current with the path energized). (Caveat: the DC test has no switching ripple, so a charger interaction that needs the boost's switching can't be 100% excluded — but it's now low-probability. `BT_SEQUENCE` has nonetheless never been tested OFF — see Next steps.)
- **`RC-BT` compensation delta (was the leading hypothesis)** — REFUTED by Death 4. `RC-BT` reverted to 61.2 kΩ (matched to FC) and the boost still died on bus-connect. Compensation `RC` is not the (sole)

cause.

- **0 V hot-plug / bring-up sequence** — REFUTED. Death 4 used `G`, which energizes the bus switches first, so the boost soft-started into a bus pre-charged to ~7.7 V. Gentle, pre-charged bring-up still kills it.
- **The boost part itself / desolder damage** — REFUTED by Death 4. The part that died was the FC channel's *known-good* TPS61288, which had just regulated 17.5 V and driven the bus on the FC pad; it also regulated 17.5 V standalone on the BT pad after reflow. It died only when driving the bus **from the BT channel**.

LEADING HYPOTHESIS: BT output-cap hot-loop inductance → destructive SW overshoot driving the bus

A known-good boost lives on the FC pad and dies on the BT pad. Schematic proven symmetric (below) → the difference is the **PCB layout** (solid by elimination). The leading specific feature is the **output-capacitor hot loop** — the pulsed, high-di/dt loop (Cout → IC VOUT/PGND → return) that sets the SW/VOUT overshoot. **All power nets here are polygon pours; trace-length figures are irrelevant (pour-overridden).** The metric that matters is **Cout distance to the IC output pin**, measured directly by the operator:

Feature	FC	BT	Note
Cout → TPS61288 output pin (operator-measured)	**40 mil (~1.0 mm)**	**240 mil (~6.1 mm)**	**6× farther on BT.** Dominant hot-loop dimension → BT hot loop has substantially higher inductance. **Leading cause.**
`VSW` inductor pad → IC pad (operator edge-to-edge)	7.4 mm	11.8 mm (1.6×)	SW is a ~150 mil-wide **pour** (low-L) carrying *continuous* inductor current (low di/dt) → minor contributor, not dominant.

Why the earlier Gerber numbers were wrong (do not reuse them): a prior parse reported `VSW-BT` "17.45 mm / 2.2×" and `VOUT`/`VBUS` trace lengths. Those summed thin stroked trace draws and **excluded the polygon pours** (`G36`/`G37` regions) that are the actual wide copper for SW *and* VOUT. They are superseded by the operator's direct pad-to-pin measurements above. Trace length ≠ loop inductance for pours.

Mechanism: BT's output caps sit 6× farther (240 vs 40 mil) from the IC output pin → larger hot-loop area/inductance → at the di/dt of driving the bus, SW/VOUT rings past the **20 V abs-max** (only ~0.5–1.5 V over the 17.5 V rail) → sync-rect FET fails **short** (`VIN`–`SW`–`VOUT`→GND, the observed post-mortem). Energy is internal $\frac{1}{2} \cdot L \cdot di^2$, so **no input current limit bounds it** (death at 120 mA). Fits Death 1 (original un-reworked boost — it's the copper), FC always surviving, the `RC` swap not helping, and the clean DC injection (no di/dt → no ring).

Inductance estimate (from the 10-mil-grid PCB images). Hot loop modeled as the VOUT pour (~150 mil) over the 2-layer GND return → microstrip $L' \approx \mu_0 \cdot h/w \approx$ **0.4–0.55 nH/mm** of one-way length, plus a **~1 nH common** term (cap ESL + vias, identical both channels):

	Cout→IC pin	length term	total hot loop
FC	40 mil (1.0 mm)	~0.5 nH	**~1.5 nH**
BT	240 mil (6.1 mm)	~3 nH	**~4 nH** (~2.7× FC; +2.5 nH)

Overshoot `V = L·di/dt`, with di/dt **backed out of FC's survival**: FC rides ≲2 V under abs-max → di/dt ≈ 1.3 A/ns → applied to BT: `4 nH × 1.3 A/ns ≈ 5.2 V` → **~22.7 V > 20 V abs-max** → **death.** Same di/dt, ~2.7× the loop inductance, turns FC's safe ~2 V into BT's fatal ~5 V. (Absolutes ±2–3×: di/dt, Coss, bottom-plane continuity all uncertain; the *relative* "FC at the edge, BT several-fold worse" is the robust part.)

Confidence: reasonably strong — the 6× placement / ~2.7× loop-L asymmetry is quantitatively consistent with BT crossing abs-max while FC doesn't, given the ~0.5–1.5 V headroom. Still **confirm with one scope capture** of `VSW`/`VOUT` on bus-connect (expect a >20 V spike on BT, a clean edge on FC). If BT does *not* ring past 20 V, re-open (ground-return/via or manufacturing defect). Fix: **add output ceramic right at the IC VOUT/PGND** (pulls BT's length term to ~0 → ~1–1.5 nH, FC-like, overshoot back to ~2 V) — both the test and the remedy in one move; ± an SW snubber.

Schematic diff — CONFIRMED SYMMETRIC (full BOM compare, `Scale_Car_Board_20260624.sch`)

Every boost-stage part matches FC↔BT, value-for-value and part-for-part:

Part	FC	BT	
`REG` boost	TPS61288LRQQR	TPS61288LRQQR	same
`L` inductor	HCM1A1305V3-2R2-R 2.2 µH	HCM1A1305V3-2R2-R 2.2 µH	same
`SNS`	INA253A1IPWR	INA253A1IPWR	same
`RC` comp	61.2 kΩ	~~27.4 kΩ~~→ now 61.2 kΩ	matched (still died)
`CC`, `RINJ`, `ROP1/2`, `RD1`, `RD2`, `R2`	2 nF / 53.6 k / 10 k+40.2 k / 237 k / 10 k / 10 k	identical	same
`C1`,`C3`,`C4A/B/C`,`C5`,`C6`,`CSNS`	10 µ/2.2 µ/3×22 µ/100 n/27 p/100 n	identical	same
`R1` (NOT a boost part)	27.4 kΩ	16.2 kΩ	**input-V ADC sense divider** (`FC_VOLTAGE`/`BT_VOLTAGE`); firmware scales each (`SCALE_V_FC`/`SCALE_V_BATT`). Different by design, irrelevant to the death.

Conclusion: the boost schematics are identical. The cause is therefore NOT a component/comp value — it is the **PCB layout** (see ROOT CAUSE above: BT `VSW` 2.2× longer).

Secondary checks (lower priority, do while reworking the BT channel): measure the BT output caps (3 × 22 µF) for cracks/lost capacitance; inspect the BT switch-node/output joints and vias; the `BT_SEQUENCE` tie was on in all four deaths but the DC path test was clean, so it's low-probability. None of these displaces the measured `VSW` layout asymmetry as the prime cause.

Safety rules for further bench work

- **No input current limit is proven safe.** Death 2 = 120 mA. Do not assume 0.3 A (or any value) protects a boost; the boost can demand >5 A, and output-side energy (its own ~10 mJ output cap, or reverse/overshoot) is not bounded by the input limit.
- **Power the Teensy/logic from a SEPARATE supply** for any bench test, so a boost-input current limit can't re-trigger the death-2 brownout/motorboating of the board-powered logic.
- **Do not install another TPS61288 until (a) the full FC-vs-BT channel + layout diff is done, (b) the BT boost-stage passives/caps are verified against FC, and (c) you can SCOPE the next attempt.** Four boosts have died without a single scope capture of the failure — do not spend a fifth blind. (The BT bus path itself is already proven clean; `R1` is the ADC sense divider, not a suspect.)

Next steps (leading cause = BT output-cap hot loop; confirm, then fix at board level)

We are out of TPS61288s and have killed four without ever capturing a scope trace. **The next boost must not be spent blind.**

1. Confirm the mechanism with ONE scoped boost (order several).

- Logic on a **separate, stiff** supply (never board-powered for the bring-up).
- **Scope `VSW-BT`/`VOUT-BT` from t=0**, single-shot armed past ~18 V, fast manual kill ready. Expectation: a **>20 V spike / ring** at the bus-connect current step, exceeding the 20 V abs-max. No input current limit is protective (energy is internal $½·L·di^2$) — scope + fast cutoff is the net.
- Capture `VSW-FC`/`VOUT-FC` on the working FC channel as the clean reference (smaller, faster-damped).

2. Board-level fix (firmware can't help) — collapse the BT hot loop first.

- **Add a ceramic output cap (e.g. 1–4.7 µF, ≥25 V) bodged RIGHT at the BT IC `VOUT`/`PGND` pins** — pull Cout from 240 mil back to FC-like (~40 mil). This is the direct fix for the measured 6× hot-loop asymmetry and the first thing to try.
- **± RC snubber across the BT SW node** (SW→GND, a few hundred pF + ~1–10 Ω, value from the scoped ring frequency) to damp residual overshoot.
- **Respin the BT channel** to place the output caps tight to the IC like FC (proper fix); also widen the `VBUS-BT` run.
- Re-derate switching current / lower bus-connect di/dt as an interim de-risk.

3. While reworking, knock out the cheap secondary checks: measure the BT output caps for cracks/lost capacitance; reflow/inspect the BT IC `VOUT`/`PGND` joints and vias; optionally run the no-boost DC-injection test with `BT_SEQUENCE` OFF to close that loose end.

Decisive confirmation: a new boost on the BT channel **with a cap added at the IC VOUT/PGND** that now survives the bus connect, scope showing the overshoot pulled under 20 V, closes this out.

Firmware status (context)

The firmware changes made across these sessions (boosts default OFF; `doState0()` gentle bring-up

- V_bus gate; `BENCH_TEST` bypass that boots to Idle with the power stage off; State-98 `G` bring-up + `1`/`2` hot-plug guard; Finish leaves bus energized) are **defensive and still reasonable**, but they target the *bring-up sequence / supply* — which the data shows is **not** the root cause. **The leading hypothesis is a BT-channel layout asymmetry (output-cap hot loop, and to a lesser degree the SW pour) causing a destructive SW overshoot, pending scope confirmation; firmware cannot fix it — it needs an SW snubber / added local `Cout` near the IC / a BT-channel respin.** The in-code comments are deliberately framed as "mechanism unconfirmed, pending scope" — treat this doc as the authoritative, current understanding.

All 219 (production) + 5 (bench) host-native tests pass: `cd test && mingw32-make`.